BMC Sports Sci Med Rehabil

https://doi.org/10.1186/s13102-026-01549-4

Article in Press

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Received: 11 October 2025

Accepted: 14 January 2026

Published online: 05 February 2026

Cite this article as: Jiang T., Zhang M., Wang Z. *et al.* The effects of integrative neuromuscular training on injury risk and athletic performance in athletes: a systematic review and meta-analysis. *BMC Sports Sci Med Rehabil* (2026). https://doi.org/10.1186/s13102-026-01549-4

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The Effects of Integrative Neuromuscular Training on Injury Risk and Athletic Performance in Athletes: A Systematic Review and Meta-Analysis

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Abstract

Background: Integrative Neuromuscular Training (INT) is a multi-component training approach aimed at optimizing neuromuscular function, improving athletic performance, and reducing injury-related risks. While existing studies suggest that INT can improve injury risk factors and enhance performance, its effectiveness and underlying mechanisms remain contentious, with many existing studies lacking comprehensive quantitative synthesis and subgroup analyses.

Methods:A systematic search was conducted across nine databases from their inception to August 31, 2025, for randomized controlled trials (RCTs) on the application of INT in athletes. The Cochrane Risk of Bias Assessment Tool (ROB2) was used for quality assessment. Meta-analysis was performed using Stata/MP 17.0, with risk ratios (RR) for dichotomous outcomes and standardized mean differences (SMD) for continuous outcomes. Hedges' g and 95% confidence intervals (CI) were calculated for effect sizes, and publication bias was assessed using funnel plots and Begg's and Egger's tests. The GRADE method was employed to assess the quality of evidence.

Results: A total of 25 RCTs (10,124 participants) were included. The meta-analysis revealed that INT is associated with a reduced injury risk in athletes (RR = 0.73, 95% CI: 0.58 – 0.91, p = 0.004), as well as improvements in jump performance (SMD = 0.88, 95% CI: 0.58 – 1.17, p < 0.001), sprinting ability (SMD = 0.75, 95% CI: 0.44 – 1.06, p < 0.001), change-of-direction ability (SMD = 0.97, 95% CI: 0.38 – 1.57, p = 0.001), and

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dynamic balance (SMD = 0.91, 95% CI: 0.31 – 1.51, p = 0.003), However, significant heterogeneity was found in most studies, and according to ROB2, the majority of studies were categorized as "having some concerns". Therefore, the certainty of the evidence should be interpreted in the context of the GRADE framework. Subgroup analysis suggests that INT may be particularly effective in adult athletes (⩾18 years), female athletes, football players, and those undergoing high-frequency (⩾3 sessions/week) and longer-duration (single session >30 minutes) interventions.

Conclusion: Current evidence suggests that INT is associated with reduced injury risk and improvements in jump performance, sprinting, change-of-direction, and dynamic balance in athletes. Adult athletes, female athletes, and football players may benefit more significantly from INT, especially when the intervention frequency is ⩾3 sessions per week and the duration exceeds 30 minutes per session.

This study has been prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO) (registration number: CRD420251131811) and was reported in strict adherence to the PRISMA 2020 guidelines. The complete PRISMA flowchart and relevant details can be found in the supplementary materials.

Keywords Integrative Neuromuscular Training ; athlete ; injury prevention ; sport performance ; Meta-Analysis

Background

Integrative Neuromuscular Training (INT) is a multi-component training paradigm with well-defined core elements, originally developed for injury prevention and performance enhancement in adolescent athletes. INT is operationally defined as a structured training programme focused on improving neuromuscular control, integrating at least two key elements within the same regimen (strength training, plyometric training, balance training, agility training, core stability). The programme is implemented in a structured and continuous manner (either as a warm-up module or as a standalone training session), with a duration of at least 15 minutes per session, and is distinct from single-component or routine warm-up interventions. Jump performance typically refers to an athlete's output during vertical or horizontal jumping tasks and is commonly quantified by jump height or distance and related metrics. Change-of-direction ability refers to an athlete's capacity to rapidly execute deceleration, directional change, and reacceleration while maintaining movement control and efficiency. INT combines general functional training with sport-specific fitness elements to optimize the neuromuscular system, enhancing athletic performance while reducing injury-related risks0[2]. Due to its comprehensive and diverse

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training methods, INT aligns with modern trends in physical conditioning and complements the ongoing pursuit of"higher, faster, stronger"in competitive sports. Consequently, INT has found wide application in both athletic performance and rehabilitation settings[3][4][5].

Compared to traditional training methods, INT places a greater emphasis on improving athletes' proprioception[6], neuromuscular[7] control[8], dynamic balance[9], agility[10], and speed within their specific sports. International research suggests that well-structured INT programs can effectively improve neuromuscular risk factors associated with sports injuries[11][12]. However, the actual effectiveness of INT and its physiological mechanisms remain subjects of debate. Some scholars argue that INT serves merely as a simple warm-up and may not lead to sustained injury prevention effects[13]. Several reviews and meta-analyses have addressed these issues: Sañudo et al.[2] conducted a systematic review focusing on adolescent athletes, confirming a beneficial trend in injury prevention and performance optimization, though they did not perform a quantitative synthesis. Chen et al.[14]'s meta-analysis focused only on performance improvements, excluding injury outcomes. Wan et al.[15] compared INT with traditional physical training but did not explore the sources of heterogeneity or subgroup differences in depth.

In summary, there are still three key limitations in the existing evidence: First, previous studies have predominantly focused on single sports or individual outcomes (injury alone or only jumping/sprinting), making it difficult to establish a unified framework that integrates "injury and athletic performance." Second, many previous reviews have treated INT as a broad concept without distinguishing the specific components of the training programme, which directly impacts the interpretation and generalization of training dosage. Third, evidence regarding training dosage (frequency/duration/cycles) and key subgroups (age, sex,athlete type) remains inconsistent[16][17]. Based on these gaps, this study aims to provide a comprehensive synthesis of the latest randomized controlled trials, incorporating both "injury risk and four athletic performance outcomes" under a unified evidence framework. It also explores dosage and population subgroups to help explain heterogeneity and enhance practical applicability, ultimately addressing the structural gaps in the current evidence and providing scientific support for individualized training programmes for athletes and coaches.

Methods

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Aim

The aim of this study is to systematically evaluate the impact of Integrative Neuromuscular Training (INT) on injury risk and key athletic performance indicators (including jump performance, sprinting, change-of-direction ability, and dynamic balance) in athletes. Additionally, subgroup analyses will explore the moderating effects of factors such as age, sex, sport type, and training dosage on the outcomes.

Literature Search Strategy

We conducted computerized searches across nine databases—CNKI, WanFang Data, VIP Database for Chinese Technical Periodicals (VIP), Cochrane Library, PubMed, Web of Science, Embase, ProQuest, and Scopus—to identify randomized controlled trials (RCTs) pertinent to our research objectives. The search period for all databases spanned from their inception to August 31, 2025, with no language restrictions. The following search strategy was employed: Population ("athlete*" OR "player*" OR "team sport*" OR "elite sport*" OR "youth sport*" OR "adolescent*"), and Intervention ("Integrative Neuromuscular Training" OR "INT" OR "NMT" OR "Neuromuscular Training"), and Outcomes("injury prevention" OR "risk of injury" OR "athletic performance" OR "sport performance" OR "physical performance" OR "jump height" OR "jump performance"OR "jump ability" OR "sprint time" OR "sprint ability" OR "sprint performance" OR "change-of-direction" OR "muscle strength" OR "balance"), and Study Type ("randomised controlled trial" OR "RCT" OR "RCTs").The searches adopted a combination of subject terms and free-text words, adjusted according to the characteristics of each database.In addition to the aforementioned searches, supplementary searches were performed on the reference lists of relevant literature, citation tracking, and key reviews, which enhanced the comprehensiveness of this review.

Inclusion and Exclusion Criteria

Inclusion Criteria:1) randomized controlled trials (RCTs); 2)Participants were athletes; 3)The experimental group followed a training programme that integrated at least two key components (such as strength training, plyometric training, balance training, agility training, and core stability) within the same regimen, implemented in a structured and continuous manner (either as a warm-up module or as a standalone training session). The control group engaged in conventional training or warm-up routines, or training without any INT components; 4)Outcome measures included at least one of

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the following: injury rate, jump performance, sprint performance, change-of-direction ability, and dynamic balance. Exclusion Criteria:1) Studies that were published multiple times; 2)Studies for which the full text or complete data could not be accessed; 3)Studies with irrelevant outcome measures; 4)Reviews or conference abstracts.

Literature Screening and Data Extraction

After retrieving the relevant literature, we imported the studies into Endnote X20 software to remove duplicates. Two researchers screened the titles and abstracts independently and in duplicate and extracted data according to a pre-designed form. In cases of disagreement, a third researcher was involved in discussions to determine whether the study should be included. Extracted data included basic information of the included studies (first author name, publication year, country), baseline characteristics of participants (age, sex, athlete type, sample size), details of the intervention (type of intervention program, duration of each intervention, intervention frequency, intervention duration), key elements of bias risk assessment, and data for the outcome measures.When a single study reports multiple follow-up time points, this study will extract data from the endpoint of the intervention. If multiple measures exist for the same outcome, we will predefine a priority order and select the most relevant indicator to avoid duplicate inclusion of data from the same study and minimize the risk of selective extraction.

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Bias Risk and Quality of Evidence Assessment

Two researchers independently assessed the risk of bias for the included studies and cross-checked the results. The risk of bias was evaluated using the Cochrane Collaboration's ROB2 tool[18]. The quality of evidence for each outcome measure was rated using the GRADE system, classifying evidence into four levels: high, moderate, low, and very low. RCTs are assumed to have high-quality evidence by default, and the quality was downgraded based on five factors: risk of bias, inconsistency, indirectness, imprecision, and other biases[19].

Statistical Analysis

Meta-analysis was performed using Stata/MP 17.0 software, with a significance level set at $P < 0.05$. The effect size combination methods were as follows: For dichotomous variables, the risk ratio (RR) and 95% confidence intervals (CI) were used, with continuity correction of 0.5 applied to

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zero-event studies,"Injury occurrence" was considered the event, and an RR < 1 indicates a reduced risk. For continuous variables, standardized mean differences (SMD) and 95% CI were calculated, using Hedges' g method to correct for small sample bias. All SMDs were coded as positive values to represent beneficial effects (e.g., shorter sprint times, higher jump heights).Study heterogeneity was assessed using the Q test ($P < 0.10$ indicating heterogeneity) and the I^2 statistic. If $I^2 \leqslant 50\%$, a fixed-effect model was applied; if $I^2 > 50\%$, a random-effects model was used to combine the effect sizes.

To explore sources of heterogeneity in the primary outcomes, we conducted prespecified stratified subgroup meta-analyses, pooling effect sizes separately by age, sex, athlete type, and training dose, and performed descriptive comparisons of pooled estimates and within-subgroup I^2 values. Because the number of studies within each subgroup was limited, we did not perform formal between-subgroup difference testing (interaction tests) and did not conduct meta-regression. Accordingly, subgroup findings were interpreted as exploratory, hypothesis-generating signals of potential effect modification rather than definitive evidence of between-subgroup differences. We also performed sensitivity analyses using a leave-one-out approach to assess the robustness of the results.

Publication bias was assessed using funnel plots, Begg's test, and Egger's test, with a statistical significance threshold of $P = 0.05$ (two-sided test). If $P > 0.05$, there was no significant publication bias; if $P < 0.05$, publication bias was present, and trim-and-fill methods were applied to adjust for it.

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Results

Literature Search and Screening

A total of 8,899 articles were retrieved through database searches. After importing the records into EndNote X20 and removing duplicates both automatically and manually, 4,221 duplicate articles were excluded, leaving 4,678 articles. Following an initial screening based on titles and abstracts, 4,557 irrelevant articles were excluded. A total of 121 articles remained, and after full-text reviews, 96 more articles were excluded. Ultimately, 25 RCTs were included in the meta-analysis (Figure 1).

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Characteristics of Included Studies

A total of 25 randomized controlled trials meeting the inclusion criteria were included in the meta-analysis, with a total of 10,124 participants from 15 countries. Of these, 5,179 participants were in the experimental group, and 4,945 were in the control group. The studies covered various sports abilities and injury-related outcomes, including: injury occurrence (8 studies), jump performance (15 studies), sprint performance (13 studies), change-of-direction ability (9 studies), and dynamic balance ability (6 studies). In all studies, the experimental group followed a training programme that adhered to the operational definition of INT or engaged in independent training sessions, while the control group (CG) used a general warm-up program. The intervention periods ranged from 4 to 30 weeks, with frequencies of 2 to 4 sessions per week and session durations between 15 and 130 minutes. Table 1 provides the basic characteristics of the included studies

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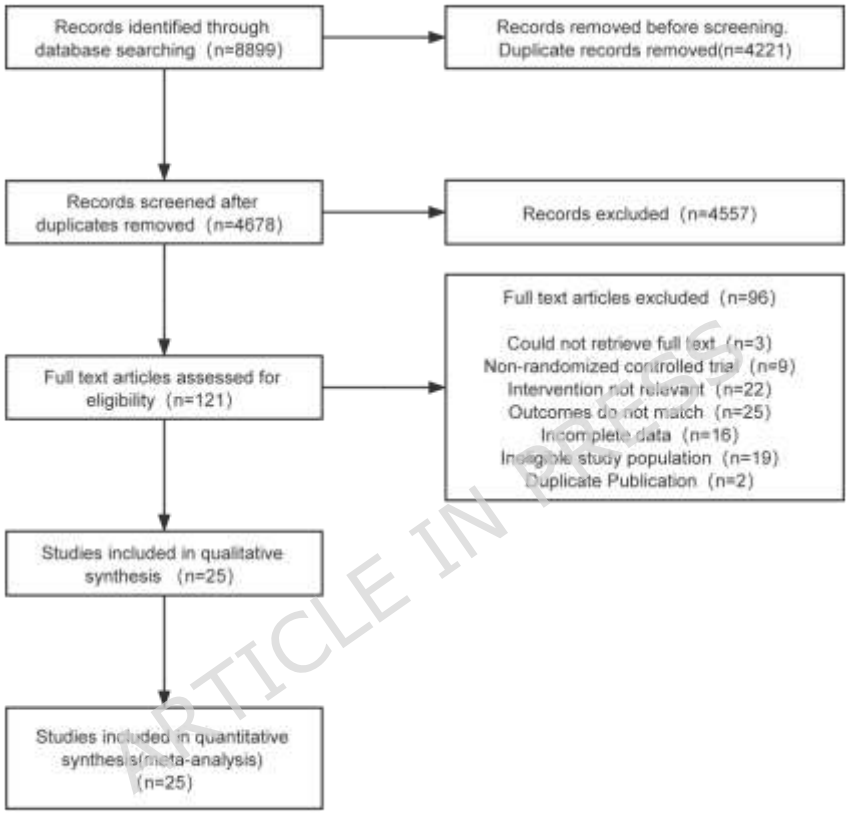

Figure 1 Literature Screening Flow Diagram

Table 1 Basic Information of Included Studies

Researcher	Nation	Research objects				Intervention measures				Outcome Measures
		sex	Type	Age/years	Sample size	Warm-up type	time/ min	frequency/ time/week	cycle/ weeks	
Shui et al.2018 [4]	China	Female	Football player	IG:17.43±0.96 CG:17.45±0.83	IG:9 CG:9	IG:Plyometric training , Strength training , Core stability training ,	90	3	6	□□□

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						Sprint training , Agility training CG:General Warm-up				
Zhao et al.2017 [5]	China	Female	Tennis player	IG:16.25±1.53 CG:16.14±1.66	IG:9 CG:9	IG:Lower limb strength training , Core stability training , Fundamental movement performance training CG:General Warm-up	90	3	8	□□
Pasanen et al.2008 [20]	Finland	Female	Cricket player	24	IG:256 CG:201	IG:Strength training,Explosive training,Sport-specific strength training,Dynamic balance training CG:General Warm-up	30	3	26	□
Emery et al.2012 [21]	Canada	Male	Football player	13—18	IG:380 CG:364	IG:Running technique training , Balance training , Strength training CG:General Warm-up	40	3	6	□
Waldén et al.2012 [22]	Sweden	Female	Football player	IG:14±1.2 CG:14.1±1.2	IG:2479 CG:2085	IG:Strength training , Agility training , Balance training CG:General Warm-up	15	2	30	□
Finch et al.2015 [23]	Australia	Male	Football player	＞18	IG:679 CG:885	IG:Knee joint control training , Core stability training , Jump training CG:General Warm-up	20	2	26	□
Benis et al.2016 [24]	Italy	Female	Basketball player	IG:20±2 CG:20±1	IG:14 CG:14	IG:Sprint training , Agility training , Jump training CG:General Warm-up	20	2	8	□
Marshall et al.2016 [25]	Canada	Female	Football player	11—18	IG:380 CG:364	IG:Static stretching , Jump training , Lower limb strength training , Core strength training CG:General Warm-up	15	2	26	□
Bonato et al.2017 [26]	Italy	Female	Basketball player	IG:20±2 CG:20±1	IG:86 CG:74	IG:Strength training , Agility training , Balance training , Aerobic training , Dynamic stretching CG:General Warm-up	120	4	30	□□□
Gee et al.2019 [27]	Britain	Female	Track and field athlete	22.3±2.0	IG:10 CG:10	IG:Strength training , Agility training , Jump training , Sprint training CG:General Warm-up	35	2	8	□□□
Lee et al.2020 [28]	Malaysia	Female	Football player	18—24	IG:5 CG:5	IG:Sprint training , Balance training , Jump training CG:General Warm-up	45—60	3	6	□

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Traković et al.2020 [29]	Serbia	Female	Volleyball player	IG:11.12±0.68 CG:10.96±0.75	IG:32 CG:34	IG:Balance training , Jump training , Core stability training CG:General Warm-up	40	2	9	□□□
Hilska et al.2021 [30]	Finland	Male	Football player	9—14	IG:673 CG:730	IG:Strength training , Jump training , Core stability training , Agility training CG:General Warm-up	20	3	20	□
Mende et al.2021 [31]	New Zealand	Female	Track and field athlete	IG:15.0±2.7 CG:15.3±2.1	IG:11 CG:11	IG:Strength training , Sprint training , Jump training , Balance training CG:General Warm-up	30	3	6	□□
Zhang et al.2021 [32]	China	Female	Dance athlete	IG:19.81±1.72 CG:20.8±1.24	IG:22 CG:20	IG:Balance training , Jump training , Sprint training CG:General Warm-up	60	3	10	□
Belamjahad et al.2024 [33]	France	Female	Football player	IG:17±1.3 CG:17±1.6	IG:12 CG:12	IG:Balance training , Sprint training , Strength training CG:General Warm-up	45-60	3	6	□□□
Belamjahad et al.2025 [34]	France	Male	Football player	IG:14.8±0.42 CG:14.7±0.56	IG:20 CG:20	IG:Interval training , Endurance training , Static stretching CG:General Warm-up	30	2	4	□□□□
Hammami et al.2025 [35]	Tunisia	Male	Football player	IG:12.4 ± 0.6 CG:12.2 ± 0.7	IG:16 CG:15	IG:Sprint training , Stability training , Jump training CG:General Warm-up	15	2	8	□□
Mainer et al.2024 [36]	Spain	Male	Tennis player	IG:13.6±0.29 CG:13.2±0.31	IG:6 CG:6	IG:Balance training , Strength training , Sprint training , Agility training CG:General Warm-up	30	2	10	□□
Xiong et al.2022 [37]	Brazil	Female	Volleyball player	IG:23.5±2.1 CG:22.9±2.4	IG:12 CG:12	IG:Strength training , Sprint training , Jump training CG:General Warm-up	30	4	8	□□
Nunes et al.2019 [38]	Brazil	Male	Volleyball player	IG:12.8±0.7 CG:13.1±0.4	IG:16 CG:16	IG:Strength training , Sprint training , Jump training CG:General Warm-up	60	2	12	□
Hammami et al.2022 [39]	Tunisia	Male	Football player	IG:15.73±0.69 CG:15.45±0.88	IG:12 CG:12	IG:Strength training , Jump training , Balance training CG:General Warm-up	15	2	8	□□□
Michailidis et al.2024	Greece	Male	Football	IG:14.2±0.6	IG:17	IG:Balance training , Strength	25	2	5	□□□

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[40]			player	CG:14.2±0.6	CG:14	training , Plyometric training , Agility training CG:General Warm-up				
Panagoulis et al.2018 [41]	Greece	Male	Football player	IG:11.2±0.5 CG:11.4±0.57	IG:14 CG:14	IG:Squat training , Jump training , Strength training , Agility training CG:General Warm-up	20	3	8	□□□
He et al.2017 [42]	China	Female	Football player	IG:23.82±1.36 CG:23.8±1.27	IG:9 CG:9	IG:Core stability training , Strength training , Agility training , Static stretching CG:General Warm-up	130	4	6	□□□

Note IG: Intervention Group; CG: Control Group □ Injury □ Jumping Ability □ Sprinting Ability □ Change-of-Direction Ability □ Dynamic Balance

Risk of Bias Evaluation

The risk of bias for the included studies was assessed using the Cochrane Risk of Bias (ROB2) tool (2019 version). This tool evaluates the risk of bias in the following five domains for randomized controlled trials: (1) randomization process; (2) deviations from the intended interventions; (3) missing outcome data; (4) outcome measurement; (5) selective reporting. Each domain was categorized as "low risk," "some concerns," or "high risk" based on identified issues. The overall risk of bias for each study was then rated.

The results of the bias risk evaluation for the 25 studies (Figure 2) indicated that 12% (n = 3) had low overall bias risk, while 88% (n = 22) had some concerns, primarily due to insufficient reporting of the randomization process and allocation concealment methods. Additionally, it was challenging to implement blinding in INT interventions. Although all studies were RCTs, the relatively high proportion of studies with concerns about bias risk suggests that the practical reference value of the findings should be evaluated in conjunction with the quality of the evidence.

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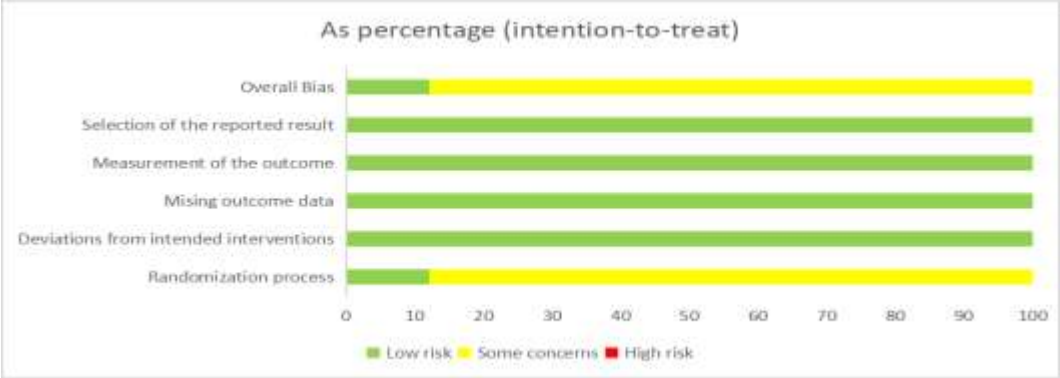

Figure 2 Risk of Bias Evaluation Results for Included Studies

Impact of INT on Athlete Injury Risk

Eight studies assessed the effect of INT on injury risk in athletes. Meta-analysis revealed significant heterogeneity between studies ($I^2 = 84.1\%$), and therefore, a random-effects model was used for the analysis. The overall result indicated that INT was associated with a reduced injury risk in athletes (RR = 0.73, 95% CI: 0.58 – 0.91, p = 0.004) (Figure 3). Sensitivity analysis, conducted by sequentially excluding individual studies (Figure 4), showed that excluding any single study did not substantively alter the direction or significance of the combined effect size (combined RR range approximately 0.58 – 0.90) .

Given the substantial heterogeneity in injury risk, we conducted prespecified exploratory subgroup analyses stratified by age, sex, athlete type, and training dose. However, because we did not perform formal tests for subgroup differences (interaction tests) or meta-regression, these results should not be interpreted as evidence of statistically significant between-subgroup differences or as definitive sources of heterogeneity.Subgroup analysis (Table 2) suggested that INT was associated with a reduced injury risk in the following subgroups: athletes aged ⩾18 years, female athletes, football players, those with an intervention frequency of ⩾3 sessions/week, and those with a session duration >30 minutes ($P < 0.05$). No significant associations were observed in subgroups of athletes aged <18 years, male athletes, non-football athletes, and those with shorter intervention durations ($P > 0.05$).It is important to emphasize that subgroup differences may arise from variations in injury characteristics and load structures across different

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sports, inconsistencies in outcome definitions, and differences in the composition and adherence to INT protocols. Therefore, this study uses subgroup results to suggest potential effect modifiers rather than to directly form definitive stratified recommendations.

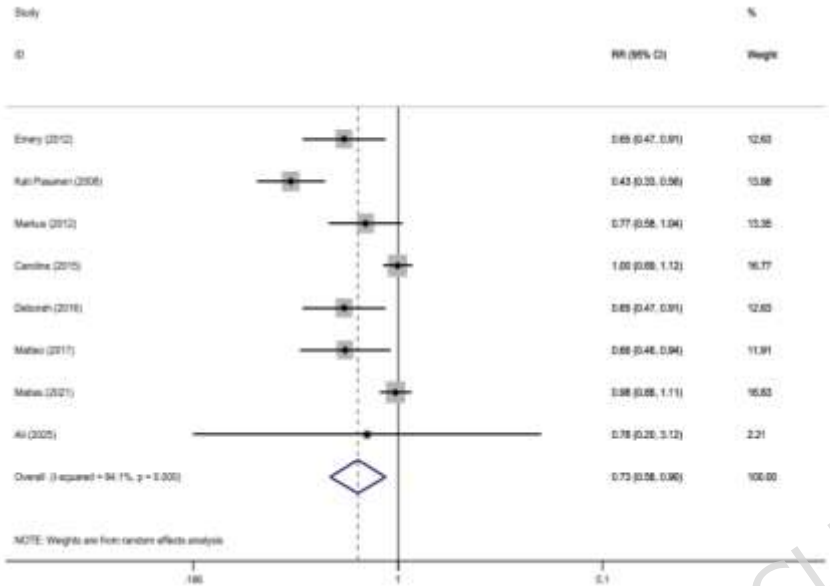

Figure 3. Forest Plot of the Effect of INT on Athlete Injuries

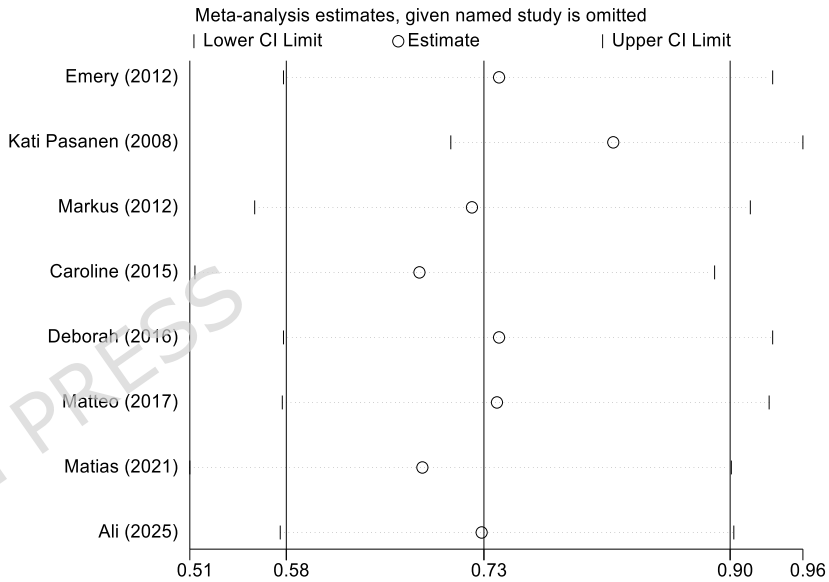

Figure 4. Sensitivity Analysis of the Effect of INT on Athlete Injuries

Table 2. Subgroup Analysis of the Effect of INT on Athlete Injuries

	Outcome Measures	Number of included studies (Number of samples)	I^2/%	Results of Meta-analysis	
				RR（95%CI）	P
Age/years	⩾18	4	91.9	0.66(0.43,0.98)	0.048
	＜18	4	47.1	0.84(0.67,1.04)	0.106
sex	Male	5	49. 5	0.90(0.78,1.04)	0.139
	Female	3	62. 1	0.56(0.41,0.75)	＜0.001
Type	Football player	6	60.4	0.85(0.73,0.99)	0.002
	Non-football players	2	71.3	0.92(0.74,1.09)	0.088
Intervention	⩽20	4	64.3	0.78(0.58,1.04)	0.090

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cycle/weeks	＞20	4	91.4	0.69(0.46,1.03)	0.070
Intervention frequency/time/ week	⩾3	4	91.0	0.66(0.43,0.99)	0.049
	＜3	4	59.8	0.83(0.65,1.06)	0.127
Intervention time/min	⩾30	4	45.9	0.56(0.43,0.74)	＜0.001
	＜30	4	61.8	0.90(0.77,1.04)	0.149

Impact of INT on Athlete Jump Performance

A total of 15 randomized controlled trials (RCTs) assessing the impact of INT on jump performance were included. Meta-analysis showed moderate heterogeneity (I^2 = 65.5%), thus a random-effects model was used. The combined effect size analysis indicated that INT is associated with improvements in jump performance (SMD = 0.88, 95% CI: 0.58 – 1.17, $p < 0.001$) (Figure 5). Sensitivity analysis, conducted by sequentially excluding individual studies (Figure 6), showed that excluding any single study did not substantively alter the direction or significance of the combined effect size (combined SMD range approximately 0.58 – 1.17) .

Subgroup analysis results (Table 3) show that all predefined subgroups exhibited significant benefits ($p < 0.05$). Additionally, in subgroups with participants aged ⩾18 years, female athletes, non-football players, intervention frequencies ⩾3 times/week, and single-session durations >30 minutes, heterogeneity decreased to 0%, suggesting that these factors may be the main sources of overall heterogeneity. It should be emphasized that the subgroup results above were obtained through stratified pooling and should be considered exploratory. Given the limited number of studies in some subgroups, we did not perform formal tests for subgroup differences (interaction tests) or meta-regression; therefore, these findings should not be interpreted as evidence of statistically significant between-subgroup differences or as definitive sources of heterogeneity. Rather, they are presented as hypothesis-generating signals to help interpret heterogeneity, and future studies using more standardized protocols are needed to validate these observations.

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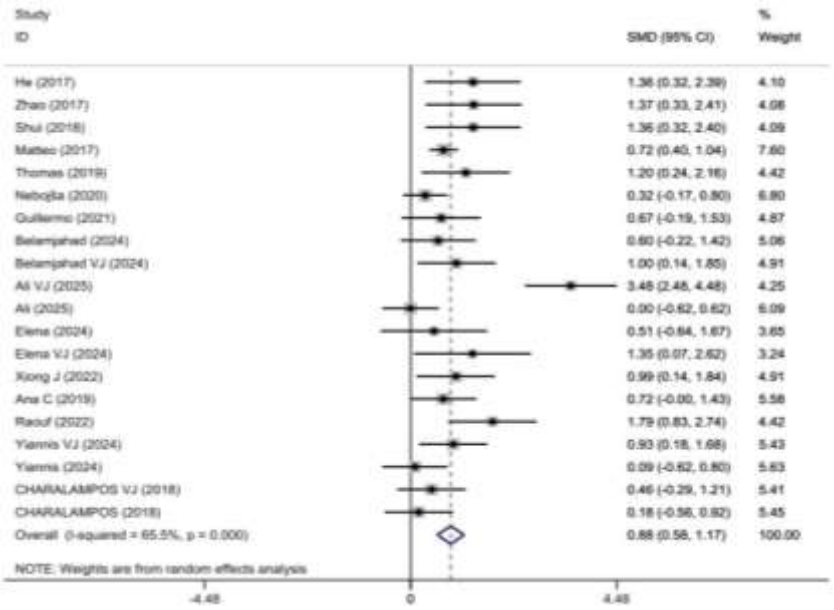

Figure 5. Forest Plot of the Effect of INT on Athlete Jumping Performance

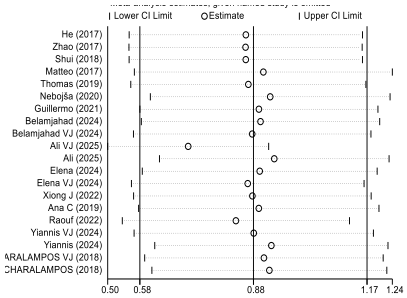

Figure 6. Sensitivity Analysis of the Effect of INT on Athlete Jumping Performance

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Table 3. Subgroup Analysis of the Effect of INT on Athlete Jumping Performance

	Outcome Measures	Number of included studies (Number of samples)	I^2/%	Results of Meta-analysis	
				SMD（95%CI）	P
Age/years	⩾18	8	0	0.88(0.64,1.12)	＜0.001
	＜18	12	77.2	0.81(0.34,1.29)	0.001
sex	Female	10	0	0.77(0.56,0.97)	＜0.001
	Male	10	80.5	0.90(0.31,1.49)	0.003
Type	Football player	11	79.3	0.97(0.43,1.51)	＜0.001
	Non-football players	9	0	0.72(0.50,0.93)	＜0.001
Intervention cycle/weeks	⩽6	9	80.4	1.01(0.38,1.63)	0.002
	＞6	11	27.6	0.75(0.49,1.01)	＜0.001

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Intervention frequency/time/ week	⩾3	9	0	0.74(0.52,0.96)	＜0.001
	＜3	11	79	0.98(0.45,1.51)	＜0.001
Intervention time/min	＞30	9	0	0.76(0.54,0.99)	＜0.001
	⩽30	11	78.5	0.90(0.36,1.45)	0.001
Outcome Measures	VJ	11	71.8	1.15(0.67,1.63)	＜0.001
	CMJ	9	45.8	0.58(0.25,0.91)	0.001

Impact of INT on Athlete Sprinting Performance

A total of 13 RCTs assessed the effect of INT on sprint performance. Meta-analysis revealed moderate heterogeneity ($I^2 = 62.4\%$), so a random-effects model was applied. The combined effect size analysis indicated that INT is associated with improvements in sprint performance (SMD = 0.75, 95% CI: 0.44 – 1.06, $p < 0.001$) (Figure 7). Sensitivity analysis, conducted by sequentially excluding individual studies(Figure 8), showed that excluding any single study did not substantively alter the direction or significance of the combined effect size (combined SMD range approximately 0.44 – 1.06).

Subgroup analysis results (Table 4) indicate that all predefined subgroups showed significant benefits ($p < 0.05$). Additionally, in the subgroups of non-football athletes and intervention frequencies <3 times/week, heterogeneity decreased to 0%, suggesting that these factors may be potential sources of heterogeneity. It should be emphasized that the subgroup results above were obtained through stratified pooling and should be considered exploratory. Given the limited number of studies in some subgroups, we did not perform formal tests for subgroup differences (interaction tests) or meta-regression; therefore, these findings should not be interpreted as evidence of statistically significant between-subgroup differences or as definitive sources of heterogeneity. Rather, they are presented as hypothesis-generating signals to help interpret heterogeneity, and future studies using more standardized protocols are needed to validate these observations.

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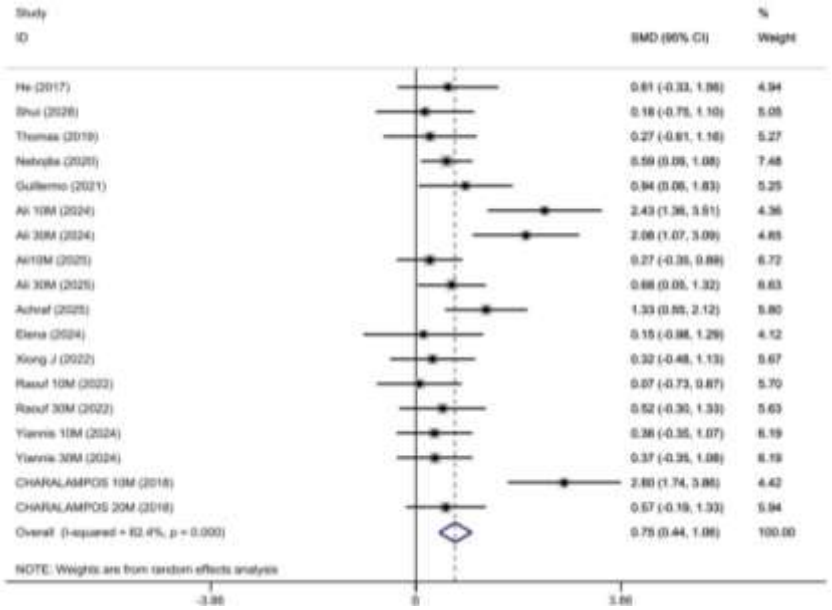

Figure 7. Forest Plot of the Effect of INT on Athlete Sprint Performance

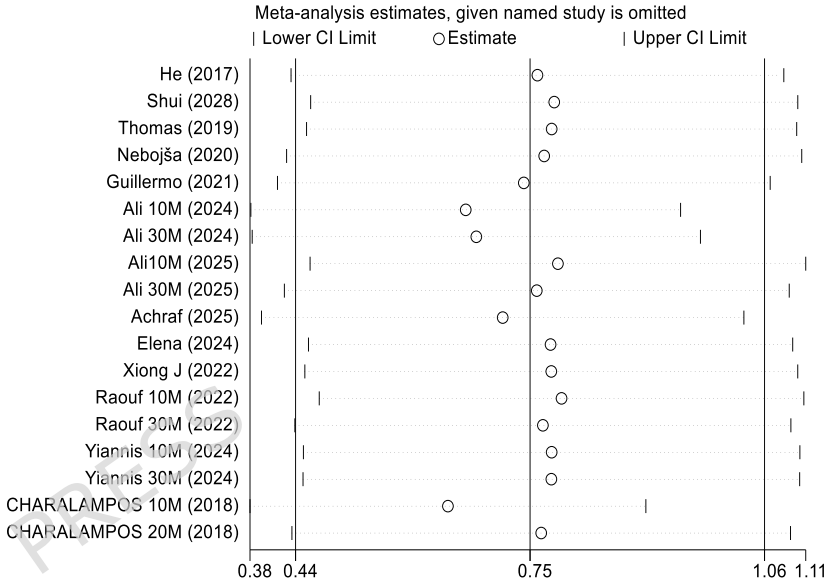

Figure 8. Sensitivity Analysis of the Effect of INT on Athlete Sprint Performance

Table 4. Subgroup Analysis of the Effect of INT on Athlete Sprint Performance

	Outcome Measures	Number of included studies (Number of samples)	I^2/%	Results of Meta-analysis	
				SMD（95%CI）	P
Age/years	⩾18	6	74.5	0.95(0.19,1.70)	0.014
	＜18	12	55.4	0.67(0.34,0.99)	＜0.001
sex	Female	8	65.7	0.87(0.36,1.38)	0.001
	Male	10	62.8	0.67(0.26,1.07)	0.001
Type	Football player	13	71.4	0.88(0.46,1.30)	＜0.001
	Non-football players	5	0	0.51(0.44,1.06)	0.003
Intervention cycle/weeks	⩽6	9	64.1	0.80(0.35,1.26)	＜0.001
	＞6	9	65.1	0.70(0.25,1.16)	0.002

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Intervention frequency/time/ week	⩾3	8	76.5	1.20(0.53,1.87)	＜0.001
	＜3	10	0	0.49(0.27,0.72)	＜0.001
Intervention time/min	⩾30	11	57	0.72(0.35,1.10)	＜0.001
	＜30	7	72.6	0.80(0.23,1.38)	0.006
Outcome Measures	30M	8	33.4	0.67(0.31,1.03)	＜0.001
	20M	4	31.8	0.65(0.12,1.17)	0.016
	10M	6	83.5	0.99(0.25,1.74)	0.009

Impact of INT on Athlete Change-of-Direction Ability

A total of 9 RCTs evaluated the effect of INT on change-of-direction ability. Meta-analysis revealed high heterogeneity (I^2 = 78.7%), so a random-effects model was used. The combined effect size analysis indicated that INT is associated with improvements in change-of-direction ability (SMD = 0.97, 95% CI: 0.38 – 1.57, p = 0.001) (Figure 9). Sensitivity analyses conducted by sequentially excluding individual studies (Figure 10) showed that removing any single study did not materially change the direction or significance of the pooled effect estimate (pooled SMD range, approximately 0.38 – 1.57).

Given the substantial heterogeneity in change-of-direction ability, we prespecified exploratory subgroup analyses by age, sex, sport, and training dose.Subgroup analysis (Table 5) indicated statistically significant improvements in change-of-direction ability for athletes aged ⩾18 years, female athletes, football players, those with an intervention period ⩽6 weeks, an intervention frequency of ⩾3 sessions/week, and those with session durations >35 minutes ($p < 0.05$). Consistent improvement effects were observed regardless of whether the IAT or T-test was used as the outcome measure ($p < 0.05$). Notably, heterogeneity decreased to 0% in the subgroups with age <18 years, non-football athletes, and intervention duration >6 weeks, suggesting that these factors may help explain the overall heterogeneity. It should be emphasized that the subgroup results above were obtained through stratified pooling and should be considered exploratory. Given the limited number of studies in some subgroups, we did not perform formal tests for subgroup differences (interaction tests) or meta-regression; therefore, these findings should not be interpreted as evidence of statistically significant between-subgroup differences or as definitive sources of heterogeneity. Rather, they are presented as hypothesis-generating signals to help

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interpret heterogeneity, and future studies using more standardized protocols are needed to validate these observations.

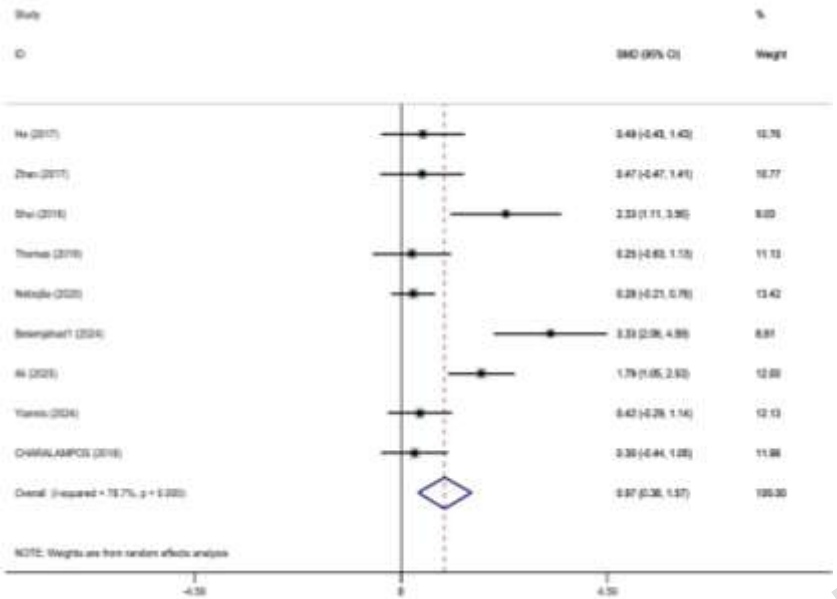

Figure 9. Forest Plot of the Effect of INT on Athlete Change-of-Direction Ability

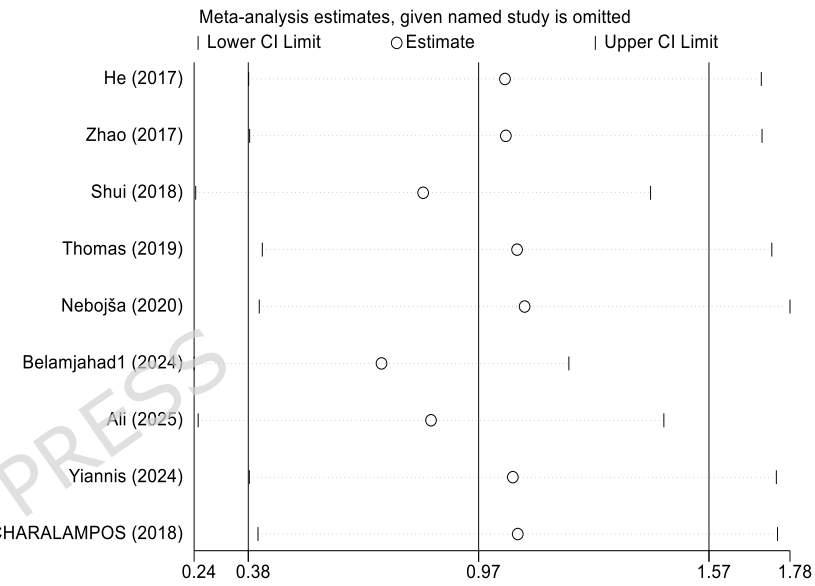

Figure 10. Sensitivity Analysis of the Effect of INT on Athlete Change-of-Direction Ability

Table 5. Subgroup Analysis of the Effect of INT on Athlete Change-of-Direction Ability

	Outcome Measures	Number of included studies (Number of samples)	I^2/%	Results of Meta-analysis	
				SMD（95%CI）	P
Age/years	⩾18	6	80.4	1.38(0.49,2.27)	0.002
	＜18	3	0	0.32(-0.03,0.67)	0.078
sex	Female	6	81.9	1.08(0.23,1.94)	0.013
	Male	3	79.2	0.84(-0.09,1.77)	0.077
Type	Football player	6	82.4	1.36(0.49,2.23)	0.002
	Non-football players	3	0	0.30(-0.08,0.69)	0.125
Intervention	⩽6	5	82.5	1.59(0.60,2.59)	0.002

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cycle/weeks	＞6	4	0	0.30(-0.04,0.65)	0.084
Intervention frequency/time/ week	⩾3	5	82.7	1.31(0.24,2.37)	0.017
	＜3	4	76	0.68(-0.03,1.38)	0.060
Intervention time/min	＞35	5	85	1.28(0.24,2.32)	0.016
	⩽35	4	72.7	0.70(-0.03,1.43)	0.061
Outcome Measures	Illinois Agility Test	5	56.1	0.64(0.05,1.24)	0.033
	T-Test	4	88.8	1.38(0.20,2.57)	0.023

Impact of INT on Athletes Dynamic Balance Performance

A total of 6 RCTs evaluated the impact of INT on dynamic balance ability. Meta-analysis revealed high heterogeneity (I^2 = 76.1%), so a random-effects model was applied. The combined effect size indicated that INT is associated with improvements in dynamic balance ability (SMD = 0.91, 95% CI: 0.31 – 1.51, p = 0.003) (Figure 11). Sensitivity analyses conducted by sequentially excluding individual studies (Figure 12) showed that removing any single study did not materially change the direction or significance of the pooled effect estimate (pooled SMD range, approximately 0.31 – 1.51).

To further investigate the substantial heterogeneity, we conducted supplementary analyses; however, owing to the limited number of eligible studies, only an exploratory subgroup analysis by athlete type was performed. The results showed that the non-football subgroup (3 studies) had no heterogeneity (I^2 = 0%) and demonstrated a small but significant effect (SMD = 0.36, 95% CI: 0.10 – 0.62, p = 0.007). In contrast, the football subgroup (3 studies) exhibited moderate heterogeneity (I^2 = 60.2%) and a larger effect (SMD = 1.81, 95% CI: 0.80 – 2.82, p < 0.01). The reduction in I^2 suggests that sport type may explain part of the heterogeneity. The difference in pooled effect sizes between the two subgroups indicates that "sport type" may be associated with the effect size. However, since no formal subgroup interaction tests or meta-regression were performed, these differences should be considered exploratory signals and require further validation through additional research.

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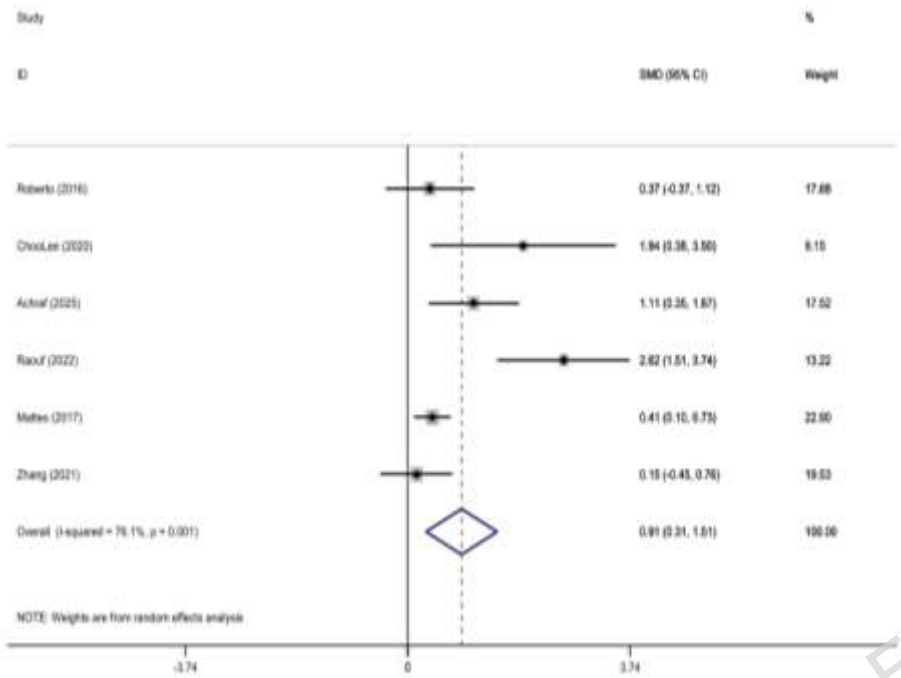

Figure 11. Forest Plot of the Effect of INT on Athlete Dynamic Balance Performance

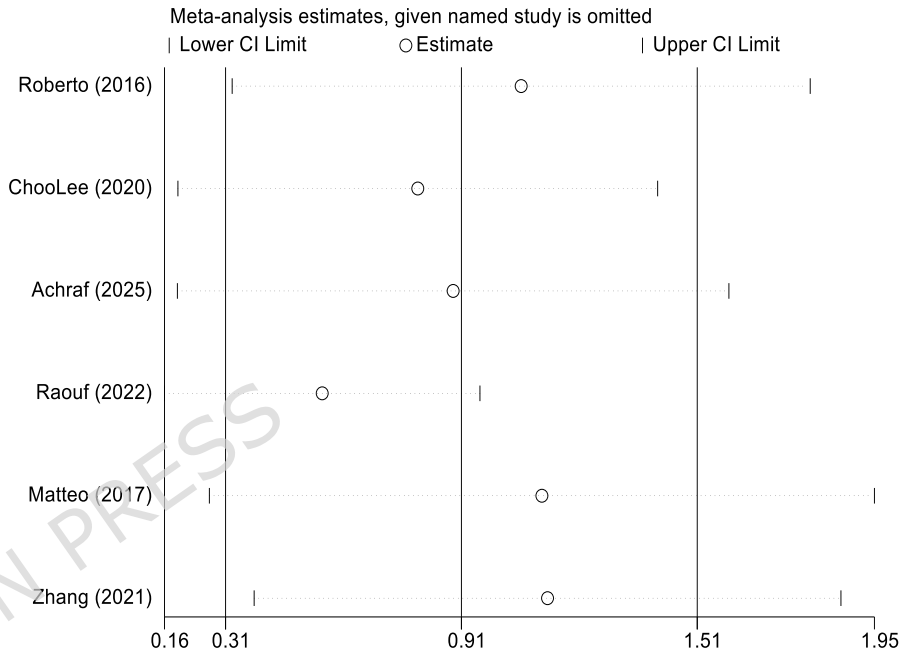

Figure 12. Sensitivity Analysis of the Effect of INT on Athlete Dynamic Balance Performance

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Publication Bias Assessment

Publication bias for the primary outcome measures was assessed using Stata 17 software (Figures 13 – 15, Table 6). Funnel plots, Egger's test, and Begg's test were used to assess publication bias for three outcome measures: jump performance, sprint performance, and change-of-direction ability. Visual inspection of the funnel plots indicated approximate symmetry, and all tests showed $p > 0.05$, suggesting no significant publication bias. No publication bias test was conducted for injury outcomes (n = 8) or dynamic balance ability (n = 6) due to the limited number of studies.

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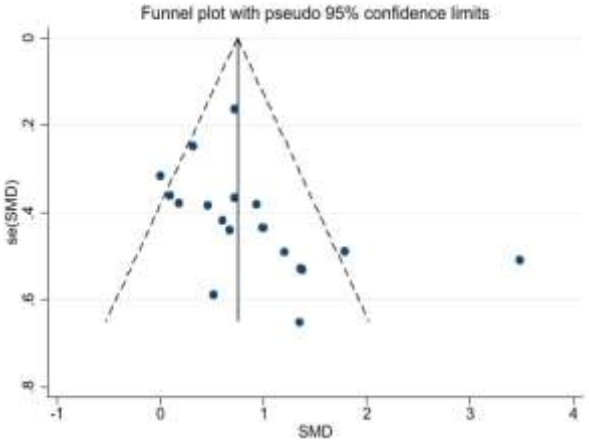

Figure 13.Funnel plot of jumping Performance

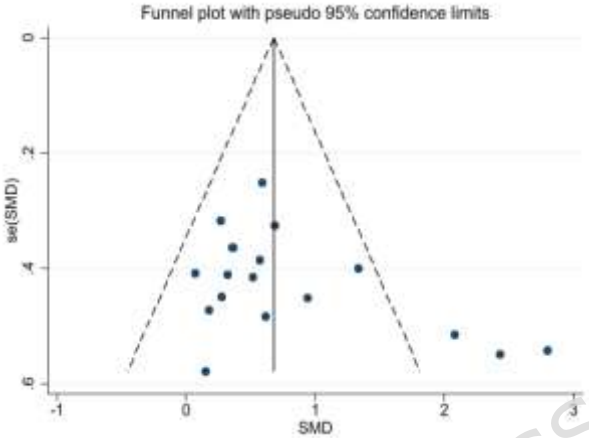

Figure 14.Funnel plot of sprinting Performance

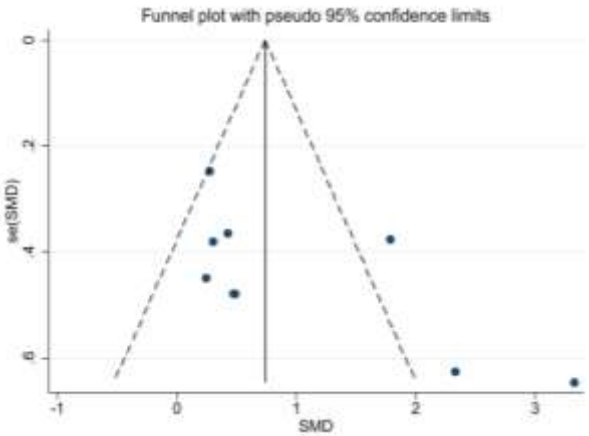

Figure 15.Funnel plot of Change-of-Direction Ability

Table 6. Summary Table of Publication Bias

Outcome Measures	The shape of the funnel plot	Egger's regression test(p)	Conclusions of bias
Jumping Performance	Symmetry	0.062	No
Sprinting Performance	Symmetry	0.075	No
Change-of-Direction Ability	Symmetry	0.070	No

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GRADE Evidence Quality Assessment

The quality of evidence for the primary outcome measures was evaluated using the GRADE method (Table 7). The results showed that jump performance and sprint performance were rated as moderate-quality evidence, while injury risk, change-of-direction ability, and dynamic balance ability were rated as low-quality evidence. The reasons for downgrading are as follows: 1) Risk of bias (all outcomes were downgraded due to missing allocation concealment in 88% of the studies); 2) Inconsistency (injury risk, change-of-direction ability, and dynamic balance ability were downgraded due to $I^2 > 60\%$); 3) Indirectness (no downgrading as INT protocols align with clinical practice); 4) Precision (no downgrading as 95% CIs did not

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include null effects); 5) Publication bias (no downgrading as tests showed no significant bias).

Table 7. Quality Assessment of Evidence for Outcome Measures

Outcome Measures	Number of studies (Articles)	Evaluation of evidence quality level					Number of interventions (persons)		Size of effect		Grade of classification
		Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	IG	CG	RR/SMD	95%CI	
Injury	8	serious[a]	serious[b]	no	no	no	4953	4723	0.73	[0.58,0.91]	Low
Jump Performance	15	serious[a]	no	no	no	no	275	262	0.88	[0.58,1.17]	Moderate
Sprint Performance	13	serious[a]	no	no	no	no	180	178	0.75	[0.44,1.06]	Moderate
Change-of-Direction Ability	9	serious[a]	serious[b]	no	no	no	132	131	0.97	[0.38,1.57]	Low
Dynamic Balance Performance	6	serious[a]	serious[b]	no	no	no	155	140	0.91	[0.31,1.51]	Low

Note a: did not implement allocation concealment; b: high heterogeneity

Discussion

This study provides a systematic review and meta-analysis to comprehensively evaluate the impact of Integrative Neuromuscular Training (INT) on athlete injury risk and various performance indicators. It is the first to conduct a subgroup analysis based on sport type. A total of 25 RCTs involving 10,124 participants were included. The results indicate that INT is associated with a reduction in injury risk and improvements in jump performance, sprinting, change-of-direction ability, and dynamic balance. These effects may be driven by shared neuromuscular adaptations that enhance both performance and injury prevention. Strength and plyometric training can enhance lower-limb force–power output and stiffness regulation, thereby improving jumping and sprint performance; greater strength may also contribute to a lower risk of injury. In addition, balance/proprioceptive and core-stability training improve postural control and joint position sense, which may reduce adverse mechanical loading during landing and cutting maneuvers. Finally, agility and change-of-direction training reinforce deceleration–reacceleration technique and neuromuscular coordination, which

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can enhance change-of-direction ability and may also reduce the risk of acute cutting-related injuries.

In comparison with previous studies, the systematic review by Sañudo et al.[2]only qualitatively summarized the beneficial trends of INT in adolescent athletes, while our study quantitatively confirmed the association between INT and reduced injury risk as well as improved multiple performance outcomes. Chen et al.[14]focused only on performance enhancement, whereas our study also provides quantitative evidence on injury outcomes and offers a more detailed examination of the differences in injury types through stratified analysis. Wan et al.[15]compared INT with traditional physical training, and our study further clarifies the target populations for INT (adult athletes, female athletes, and football players) and training parameters (⩾3 sessions/week, session duration >30 minutes).It is important to note that the findings from the subgroup analyses are exploratory, as some subgroups had a limited number of studies, preventing definitive conclusions.

We systematically evaluated the internal validity and certainty of the included evidence using ROB2 and GRADE. ROB2 assessments indicated that most studies raised "some concerns," mainly due to insufficient reporting of the randomization process and allocation concealment; accordingly, we generally downgraded the certainty of evidence by one level for risk of bias in the GRADE assessment across outcomes. Overall, the certainty of evidence was moderate for jumping and sprint performance, whereas it was low for injury outcomes, change-of-direction ability, and dynamic balance, largely because substantial between-study variability led to downgrading for inconsistency. Although tests for publication bias did not indicate clear asymmetry, statistical power is limited for outcomes with few included studies, and the influence of publication bias and imprecision cannot be fully excluded. Future preregistered RCTs with more complete methodological reporting and larger sample sizes are needed to strengthen the robustness and generalizability of these findings.

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Effect of INT on Athlete Injury Risk

A total of 8 studies focused on injury prevention were included in this analysis. The results demonstrated that INT is associated with a reduced injury risk in football players, consistent with several previous studies[2][42][43]. The primary mechanism may involve improvements in lower limb biomechanics and muscle strength balance. For instance, INT has been shown to optimize knee valgus angle and torque, improving biomechanical load

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distribution and indirectly reducing the risk of injuries such as anterior cruciate ligament (ACL) tears. Pengfei He et al.[42]observed significant improvements in knee valgus angle and torque after INT intervention in female football players, supporting the beneficial effect of INT on biomechanical load distribution. HEWETT's prospective study[44]also confirmed that lower limb mechanical abnormalities are a significant risk factor for ACL injuries, indirectly suggesting that INT may reduce injury risk by improving this factor. Additionally, other studies have proposed that neuromuscular adaptations, such as enhanced motor unit recruitment and optimized force sequencing, may also contribute to injury prevention[45][46]. However, the studies included in our analysis did not provide direct evidence for these mechanisms, and further investigation is needed.

The first sport-specific subgroup analysis indicated that INT may confer a protective benefit in football players, whereas no consistent benefit was observed in non-football athletes. This finding aligns with Castillo et al.[47]. The core mechanism likely lies in the specific adaptation of movement patterns to INT training: the high-frequency change-of-direction, sudden stops, and kicking actions in football closely resemble the functional movements in INT. Athletes can transfer the improved joint stability control and optimized force sequencing from training to competition, creating an efficient "training-to-practice" transfer[48]. In contrast, non-football sports have different movement patterns, with distinct lower limb force angles and muscle coordination, making it harder for neuromuscular adaptations to transfer effectively[49]. This reduces the intervention benefits. Our study provides supporting evidence for the sport-specific application of INT in football training.

Age and sex are also key factors influencing INT outcomes. Our study found that INT has a significant protective effect in athletes aged ⩾18 years, while no significant effects were observed in adolescent athletes (<18 years), consistent with the findings of Myer et al.[50]. The core mechanism here is that adolescent athletes are in a critical growth phase where skeletal growth outpaces muscle and ligament development, leading to increased joint laxity and mismatch between muscle strength and skeletal load, It makes it difficult to achieve the desired intervention effect. Additionally, the central nervous system's motor unit recruitment efficiency and discharge synchronization have not yet reached their peak, limiting stable neuromuscular adaptations from INT training[51]. In contrast, adult athletes have more developed neuromuscular systems and more stable physiological structures, enabling quicker correction of movement errors and optimizing muscle fiber type conversion and tendon elasticity, which

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further enhances injury prevention. sex differences in INT outcomes stem from inherent differences in lower limb biomechanics and hormone levels. INT has shown more significant injury prevention effects in female athletes, likely due to their higher knee valgus moments and relatively weaker knee stability. INT training targets muscle groups such as the gluteus medius and the medial quadriceps, which helps counteract valgus moments and improve dynamic alignment, offering significant protection from injury[52]. Male athletes, with stronger foundational muscle strength and joint stability, may experience relatively lower injury risks, so the marginal benefits of INT training are less pronounced. Therefore, more intense or prolonged interventions may be needed to observe significant effects.

Training dosage is another critical parameter affecting INT outcomes. A training frequency of⩾3 sessions per week and session duration >30 minutes ensures that the neuromuscular system receives sufficient and continuous stimulus, promoting fascial chain elasticity and central motor program remodeling, leading to stable motor pattern memory. In contrast, low-frequency and short-duration interventions may fail to meet the neuromuscular adaptation threshold, resulting in diminished training effects. This aligns with the dose-response relationship proposed by MYER et al.[50], which suggests that higher frequencies and longer durations of training provide sufficient stimulation to induce significant neuromuscular adaptations[53]. This training dosage also aligns with recommendations by SUGIMOTO et al.[54] and SOLIGARD et al.[55] for maintaining an INT frequency of more than 3 sessions per week during the preparatory period and regular training to accumulate effective physiological adaptations. Our study's dose-response findings lead to the following practical recommendations: female football players aged ⩾18 years should adopt an INT program with 3 sessions per week, each lasting 30–45 minutes, for⩾8 weeks, incorporating strength training, jump training, balance training, and agility training to maximize injury prevention effects.

Existing studies have shown that INT is effective in reducing ACL injury risk and improving knee and ankle joint stability[56][57]. However, these studies often cover a broad range of sports, which, while helpful in making general conclusions, may mask the specific effects of INT on particular sports due to heterogeneity between sports. Our study, through subgroup analysis, minimizes this heterogeneity and provides a more precise assessment of INT's injury prevention effect in football-specific injury patterns. However, due to the limited number of studies included in this analysis,

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a separate analysis for each sport was not conducted, which remains a limitation of this study. Future research incorporating more sport-specific RCTs is needed to enhance the comprehensiveness and persuasive power of the results.

Effect of INT on Athlete Performance

This study included 18 articles related to performance outcomes, demonstrating that Integrative Neuromuscular Training (INT) is associated with improvements in athletes' jump performance, sprinting, change-of-direction ability, and dynamic balance. These findings are consistent with several previous studies[58][59][60]. INT, as a comprehensive training approach, is effective in stimulating and optimizing the neuromuscular system across multiple dimensions, enhancing the stretch-shortening cycle to improve explosiveness and movement speed[61][62]. Moreover, by repeatedly practicing functional movement patterns, INT corrects poor biomechanical patterns and improves movement efficiency, allowing athletes to develop more effective movement strategies[63].

The effect of INT on change-of-direction ability showed significant subgroup differences, with the core mechanism related to the precision of neuromuscular control and the adaptability of movement patterns. Subgroup analysis revealed that INT had a more significant impact on athletes aged ⩾18 years, female athletes, and football players. In adult athletes, the central nervous system's ability to integrate and process proprioceptive signals is stronger, allowing for quicker adjustments in the timing of antagonist and synergist muscle contractions during directional changes, thus reducing energy leakage[56]. Female athletes benefit from INT training by improving the efficiency of the hip-knee-ankle kinetic chain, correcting pelvic anterior tilt and knee valgus during direction changes, and significantly enhancing direction change speed and stability. For football players, the high similarity between training movements and competition scenarios allows INT exercises, such as agility ladder drills and obstacle course direction changes, to directly enhance the "observe-decide-act" neural pathways used during games, facilitating rapid transfer of training to performance[64].

This study found that training parameters play a crucial role in modulating the effect on change-of-direction ability, with the timing of neuromuscular adaptations being a key factor. When the intervention period is ⩽6 weeks, the neuromuscular system is in the rapid adaptation phase. High-frequency (⩾3 sessions/week) training strengthens synaptic efficiency, increasing the speed of movement command transmission, while session

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durations >35 minutes ensure adequate stimulation of core muscles, optimizing muscle coordination. After 6 weeks, neuromuscular adaptations plateau, requiring adjustments in training intensity or type to further enhance effects. This also explains the disappearance of heterogeneity in the long-term subgroup, where the differences in effects are more likely due to individual physiological factors rather than training parameters[65]. Both IAT and T-test results showed significant improvements, indicating that INT's effect on change-of-direction ability is universal and not dependent on a specific test method, thus increasing the external validity of the findings.

The improvement in dynamic balance ability stems from the collaborative optimization of proprioception and posture control. INT, through exercises such as balance boards and single-leg holds, enhances the sensitivity of the skeletal muscle system around the joints, improving the central nervous system's ability to perceive and regulate shifts in body weight. At the same time, it strengthens the stabilizing role of the core muscles, forming a "perception-feedback-regulation" feedback loop that provides a stable foundation for activities such as jumping, sprinting, and direction changes[66]. The differences in effects among populations are closely related to neuromuscular maturity and the relevance of movement patterns. Adult football players, benefiting from training adaptability and physiological advantages, show the most significant improvements,Adolescent football players may have already reached certain physiological and neurological limits in the process of improving their athletic performance, which makes it difficult to observe short-term training effects. INT strengthens joint muscles, ligaments, and proprioception, enhancing muscle coordination and posture control, which improves the dynamic stability of key areas such as the knee and ankle[67]. This reduces energy leakage, optimizes force transmission in the movement chain, and helps athletes maintain better balance and control during high-speed movements, direction changes, or landing from jumps, thereby improving dynamic balance and indirectly supporting jump, sprint, and change-of-direction ability[68].

Based on these findings, specific recommendations for different populations are as follows: adult athletes can follow a program of 3 sessions per week, with each session lasting 30 – 60 minutes; adolescent athletes should start with 2 sessions per week, each lasting 20 – 30 minutes, gradually increasing the dosage; non-football athletes should adjust core components based on their sport's characteristics.

In conclusion, INT may improve athletic performance and reduce injury risk through neural adaptations, biomechanical optimization, and

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improved energy-use efficiency. The subgroup differences observed may primarily reflect inherent variation in physiological maturation, the precision of neuromuscular control, the suitability of movement patterns, and individual thresholds for training stimuli.

This study has several limitations. Injury definitions, follow-up windows, and reporting methods varied across studies; although stratified analyses partly addressed these issues, residual confounding may remain and could have inflated statistical heterogeneity for injury outcomes. Differences in INT protocols across studies may also have increased heterogeneity and limited the precision of dose – response inferences. Moreover, 88% of studies lacked allocation concealment and blinding was not feasible, which may have introduced bias. We did not assess minimal clinically important differences, and it therefore remains unclear whether the observed effect sizes translate into clinically meaningful benefits. This study's exploration of heterogeneity primarily relied on stratified subgroup meta-analyses and descriptive comparisons of within-subgroup heterogeneity. Due to the limited number of studies and insufficient data within each subgroup, we did not perform formal subgroup interaction tests or meta-regression. As such, any interpretations related to subgroup differences should be regarded as exploratory, potentially influenced by multiple comparisons and random significance, and require further validation through larger, more standardized studies in the future.

Future research should prioritize developing a reproducible minimum set of INT components and standardizing the reporting of adherence and training load; concurrently measuring potential mediators in injury-prevention trials (landing biomechanics, neuromuscular activation, and strength/power metrics) to test the mechanistic pathway; and using approaches such as stratified randomization to improve generalizability and statistical power across sports and across sex- and age-specific strata.

Conclusion

The results of this study suggest that Integrative Neuromuscular Training (INT) may be associated with a lower injury risk and improvements in multidimensional athletic performance. The available evidence indicates that these effects may be influenced by population characteristics and training dosage, with more pronounced benefits observed for some outcomes in adult athletes, female athletes, and football players; the associations appear more stable when training is delivered at a frequency of ⩾3 sessions per week and a session duration of ⩾30 minutes. Accordingly, coaches and

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practitioners may consider incorporating INT as a routine component of training, implemented at ⩾3 sessions per week for 30–45 minutes per session, with progression and individualization based on sport-specific demands and individual characteristics. However, the certainty of evidence for most outcomes ranged from low to moderate, and the subgroup analyses were exploratory. High-quality randomized controlled trials across a broader range of athletic populations are therefore needed to further confirm the applicability of INT.

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List of abbreviations

Integrative Neuromuscular Training (INT)

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Randomised Controlled Trial（RCT）

Intervention Group（IG）

Control Group（CG）

Acknowledgements

All authors contributed to the study idea and design. TC.J.wrote the ffrst draft of the manuscript. TC.J. and PF.W. and ZL.Wwere responsible for the search strategy, research selection, and data extraction. TC.J.and PF.W. and MY.Z. and ZL.Wperformed the literature assessment. All authors read and ffnalized the ffnal manuscript.

Author contributions

TC.J.wrote the main manuscript. PF.W. and ZL.W. provided research methodology and supervision. PF.W. and MY.Z. provides writing supervision and review.All authors reviewed the manuscript.

Funding

no external funding.

Data availability

Tiance, Jiang (2025), “The Effects of Integrative Neuromuscular Training on Injury Risk and Athletic Performance in Athletes: A Systematic Review and MetaAnalysis”, Mendeley Data, V1, doi: 10.17632/hddjvhvs9s.1.

Declarations

Ethics approval and consent to participate

not applicable.

Consent for publication

not applicable.

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Competing interests

The authors declare no competing interests.

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